

Chapter 7

1. Let $H = \{0, \pm 3, \pm 6, \pm 9, \dots\}$. $H = 3\mathbb{Z}$

$$0 + H = H$$

$$1 + 3\mathbb{Z}$$

$$2 + 3\mathbb{Z}$$

$$3 + 3\mathbb{Z} = H$$

Left cosets: $H, 1 + 3\mathbb{Z}, 2 + 3\mathbb{Z}$

3. $a + H = b + H$ iff $a - b \in H$

a. $11 - 17 = -6 \in H$. These cosets are the same

b. $-1 - 5 = -6 \in H$. These cosets are the same

c. $7 - 23 = -16 \notin H$. These cosets are distinct

4. $|\mathbb{Z}| / |n\mathbb{Z}| = n$

There are n distinct cosets

6. Let $|a| = 15$.

We know $\langle a \rangle = \{e, a, a^2, \dots, a^{14}\}$

$$\langle a^5 \rangle = \{e, a^5, a^{10}\}$$

Find cosets with form $g \langle a^5 \rangle$ $\forall g \in \langle a \rangle$.

$| \langle a \rangle | / | \langle a^5 \rangle | = 5$. Thus there are 5 distinct cosets.

Corresponding to $a^k \langle a^5 \rangle$ where $k < 5$

$$k=0: a^0 \langle a^5 \rangle = \langle a^5 \rangle = \{e, a^5, a^{10}\}$$

$$k=1: a^1 \langle a^5 \rangle = \{a, a^6, a^{11}\}$$

$$k=2: a^2 \langle a^5 \rangle = \{a^2, a^7, a^{12}\}$$

$$k=3: a^3 \langle a^5 \rangle = \{a^3, a^8, a^{13}\}$$

$$k=4: a^4 \langle a^5 \rangle = \{a^4, a^9, a^{14}\}$$

7. $| \langle a \rangle | / | \langle a^4 \rangle | = 30 / 30 \div \gcd(30, 4) = 30 / 15 = 2$

$$\forall k < 2$$

$$k=0: a^0 \langle a^4 \rangle = \{e, a^4, a^8, a^{12}, a^{16}, a^{20}, a^{24}, a^{28}, a^2, a^6, a^{10}, a^{14}\}$$

$$k=1: a^1 \langle a^4 \rangle = \{a, a^5, a^9, a^{13}, a^{17}, a^{21}, a^{25}, a^{29}, a^3, a^7, a^{11}, a^{15}\}$$

8. Let $G = S_3$. Given $H, K \leq G$, Let $H = \{e, (13)\}$,
 $K = \{e, (12)\}$

Show HK violates subgroup criteria

$$HK = \{e, (13), (12), (123)\}. HK \neq \emptyset$$

$$\text{closure: } (123)(123) = (132) \notin HK$$

$$\therefore HK \neq G$$

9. Let $H = \{(1), (12)(34), (13)(24), (14)(23)\}$

$$\text{Find } [A_4 : H] = 12 / 4 = 3$$

$$\text{Find } [S_4 : H] = 24 / 4 = 6$$

15. Divisors of 60: 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

19. $5^{15} \pmod{7}$

$$= (5^5)^3 \pmod{7} = (125)^3 \pmod{7} = 6^3 \pmod{7} = 7776 \pmod{7} = 6$$

$$7^{13} \pmod{11}$$

$$7^{10} \equiv 1 \pmod{11} \quad [\text{Fermat's little theorem}]$$

We know $13 = 10 + 3$, so $7^{13} = 7^{10} \cdot 7^3$

Since $7^{10} \equiv 1 \pmod{11}$, find 7^3 .

$$7^3 = 343$$

$$343 \pmod{11} = 2$$

$$7^{13} \pmod{11} = 2$$

20. Recall $|U(n)| = \phi(n)$ [Euler phi function]

$n > 2$.

$\forall a \in U(n)$, $\exists b \in U(n)$ such that $ab \equiv 1 \pmod{n}$

because $U(n)$ is group under multiplication mod n

Consider element $-1 \pmod{n} \in U(n)$. We know -1 has order 2 and by Corollary 2 $2 \mid \phi(n)$

$\therefore \phi(n)$ is even

22. Let G be a group and $H, K \leq G$ with $|H| = 12$, $|K| = 35$

$\gcd(|H|, |K|) = 1$ since $|H \cap K| / |H| \nmid |H \cap K| / |K|$

Thus $|H \cap K| = 1$

$\Rightarrow$ Let G be a group and $H, K \leq G$. If order of H and K are relatively prime, $H \cap K$ is the trivial subgroup

23. Consider $n \geq 3$, since 2 is odd

When n is even:

Consider $r^{n/2} \Rightarrow T_{180} ; e ; f$ (flip); $f \cdot T_{180}$

$K = \{e, T_{180}, f, f \cdot T_{180}\}$

In subset K , order of every element divides $|K|$ and each element is its own inverse.

$K \leq D_n$ by construction

when n is odd:

To compose a subgroup with order 4, order of every element must divide 4. Divisors of 4: 1, 2, 4. However, no rotation $4 \nmid 2n$ odd

in D_n will have order that divides n . Because of lack of rotations of order 4, condition does not hold.

25. Let G be an Abelian group and $|G|$ be odd.

We can pair each element $a \in G$ with its inverse a^{-1} .

If $a \neq a^{-1}$, we know $a a^{-1}$ is the identity.

We also know there are no self inverses since $|G|$ is odd.

Hence identity in G multiplied in series with these pair of elements in G will produce identity element, e .

26. Let G be a group and $|G| \geq 1$. [show $|G|$ is prime]

Since G has no proper subgroups, then possible subgroups must either be G itself or $\langle e \rangle$.

Thus for $g \neq e$, $\langle g \rangle = G$.

Thus $|G|$ must be prime as this corresponds to have two divisors $|\langle e \rangle|$ and $|G|$.

27. Let G be a group and $|G| = 15$.

Divisors of 15: 1, 3, 5, 15.

We know in any group G , it is its own subgroup.

By definition of group, there is also only one identity, e with order 1.

Thus, we've shown there's only one unique subgroup with order 1 and order equal to $|G|$.

Given G now has only one subgroup of order 3 and 5, for all divisors of $|G|$, there exists only one subgroup of that size.

$\therefore G$ must be cyclic.

* 28. Let G be a group and $|G| = 25$.

34. LCM(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)

$$1 = 1$$

$$2 = 2$$

$$3 = 3$$

$$4 = 2^2$$

$$5 = 5$$

$$6 = 2 \times 3$$

$$7 = 7$$

$$8 = 2^3$$

$$9 = 3^2$$

$$10 = 2 \times 5$$

Looking at highest powers: $5^1, 3^2, 2^3, 7^1$
 $= 5 \times 9 \times 8 \times 7 = 2520$

36. Let G be a group and $|G| = 21$

Divisors of 21: 1, 3, 7, 21

Given $g^{14} = e$, $|g| \mid 14$

Divisors of 14: 1, 2, 7

Like divisors are $1 \nmid 7$

$\therefore |g| = \{1, 7\}$

38.

39. Let H, K be subgroups order 24, 20 respectively
 $|H \cap K| \mid |H| \nmid |H \cap K| \mid |K|$ [Lagrange's theorem]

$$\gcd(24, 20) = 4$$

Possible orders: 1, 2, 4

Case 1:

$$|H \cap K| = 4$$

$H \cap K \cong \mathbb{Z}_4$. Thus it is Abelian

Case 2:

$|H \cap K| = 2$. $H \cap K \cong \mathbb{Z}_2$. Thus it is Abelian

Case 3:

$|H \cap K| = 1$. Thus H is the trivial subgroup which is Abelian.

$\therefore H \cap K$ must be Abelian.

42. Let G be a group and $H \leq G$ with order 100 and 25 respectively.

Observe $|H| \mid |G|$. Assume $\exists g \in G$ such that $|g| = 5$.

We know the order of every element $g \in G$ divides $|G|$ so there's an element of order 5 in group G .

Suppose for contradiction $g \notin H$. Let $K = \langle g \rangle$.

$$|K| = 5$$

Since H is a subgroup and K is a cyclic subgroup, then $H \cap K$ is a subgroup.

We know $|H \cap K| \mid |H| \nmid |K|$, so $|H \cap K| = 1, 5$.

Case 1:

$|H \cap K| = 5$ implies $K \leq H$ which contradicts assumption.

Case 2:

$|H \cap K| = 1$ implies $H \cap K$ is the trivial subgroup.

$$|HK| = |H||K| \div |H \cap K| = 25 \cdot 5 = 125$$

But $|G| = 100$ and product HK cannot have order 125.

$\therefore$ It is impossible for $g \in G$ with order 5 and $g \notin H$.
 g must be in H .

45. Let $G = \{(1), (12)(34), \dots\}$

a. $\text{stab}_G(1) = \{(24)(56), (1)\}$

$$\text{Orb}_G(1) = \{1, 2, 3, 4\}$$

b. $\text{stab}_G(3) = \{(1), (24)(56)\}$

$$\text{Orb}_G(3) = \{3, 4, 1, 2\}$$

c. $\text{stab}_G(5) = \{(1), (12)(34), (13)(24), (14)(23)\}$

$$\text{ord}(5) = \{5, 6\}$$

16. Let G be a group with order 12

Assume for contradiction, G has no elements of order 2

Then, all 11 non-identity elements have possible orders $\{3, 4, 6, 12\}$

Case 1: order = 3

Each group generated by element of order 3 has 2 elements of order 3. [Euler's function]

Case 2: order = 4

Similarly, each subgroup generated by element of order 4 has 2 elements of order 4.

Case 3: Order = 6

Likewise, there will be 2 elements of order 6 in each

Case 4: order = 12

In this case, G would be cyclic and $\exists g \in G$ such that $|g| = 2$ which contradicts our assumption

Adding up these elements show there's an even number of non-identity elements that make up the group G which contradicts $|G \setminus \{e\}| = 11$

17. Let G be a group and $H, K \leq G$ and $|H| = pq$, $|K| = qr$

$$|G| = pqr$$

$$|H \cap K| \mid |H| \quad \& \quad |H \cap K| \mid |K|$$

$$\gcd(pq, qr) = q$$

Given that there are elements in both H and K of order prime q , $H \cap K$ must share this element so

$$|H \cap K| \neq 1$$

$$\therefore |H \cap K| = q$$

50 We know $|A_5| = 60$

Consider $\text{stab}_A(5)$

$$|A_5| = |\text{Orb}_A(5)| |\text{stab}_A(5)|$$

We know in A_5 $\text{orb}_A(5) = \{1, 2, 3, 4, 5\}$

$$|\text{orb}_A(5)| = 5$$

$$|\text{stab}_A(5)| = 60 \div 5 = 12$$

Since $\text{stab}_A(5) \leq A_5$, A_5 contains subgroup of order 12